Ask Madden, but Maddon: A Fan Interactive Activity to Understand Relays

Team 225

Abstract

When it comes to thinking about winning on the field, MLB teams think in terms of run scoring and run prevention. One crucial but often overlooked component of run prevention is relays. For the casual fan, however, the impact of a relay is often misunderstood as there are many fluid components of a relay that contribute to the runner being safe or out. To enhance fan engagement in relay outcome, we constructed a random forest classification model that can predict the outcome of a relay. This model serves as "Ask Madden", which can be used in a fan interactive game that allows fans to see and explain the actions on the field that contribute to a well-executed relay.

1. Introduction

In every ballpark, there is typically an activity or a promotion done between innings. Whether it's Mr. Freeze racing somebody around the warning track or someone trying to track the ball under shuffling caps, these are very easy ways to keep fans engaged between innings. However, these activities have nothing to do with the game. It's also common to see the same game being played in other ballparks; hence, there is a lack of variety. With that in mind, the purpose of this project is to create a fan activity that not only is different from other fan activities but also helps fans understand the action on the field.

As an avid sports video game player growing up, I thought about how sports video games transformed casual fans of the sport into invested fans that started thinking about their favorite sport differently. The video game series that came to mind was the Madden NFL series. For me, "Ask Madden" seamlessly implemented football concepts and strategy into their game by not only simplifying the playbook through suggesting plays, but explaining the rationale behind the suggestion in a way that made sense to casual fans and diehards alike (see Figure 1). Thus, I knew that my fan activity would include an "Ask Madden" feature that could explain the on-field action in a fan-friendly way.



Figure 1: Ask Madden in Madden 12

When thinking about ways that a mid-game fan interactive activity can enhance fan understanding of baseball, I was thinking about aspects of the game that feel miniscule in the grand scheme of the game but are important to a team's ability to score and prevent runs. The first thing that came to mind was the relay. Throughout my high school baseball days, our team would spend most of the practice time on defensive structure. However, the number of times

where we would execute our structure was very few, so it's easy to understand why fans might tend to interpret relays in an outcome-oriented way rather than trying to understand the actions that go into the relay. An example of this thinking can be seen from my demonstration, which uses a relay from the 2024 College World Series Championship. The runner in this play makes a swim move at home, scoring what ended up being the winning run in the game (see Figure 2). Even though the relay was well-executed, it's lost on fans because they only remember how great the slide was.



Figure 2: Tennessee's Hunter Ensley makes a wild slide to avoid the tag and score the run for Tennessee in the 2024 College World Series Final.

Understanding the value of relays is crucial when it comes to thinking about run prevention. Baseball Instincts (Reference #3), a coaching blog dedicated to drill in-game situations, calls a double cut relay baseball's most "misunderstood play." Even though double cut relays are engaged when a ball is hit into the deep outfield, players and coaches understand that it's now about limiting damage rather than worrying about getting the hitter out, so it's important for players to understand what they need to do to limit their losses and stay in control of the game. Outside of the on-field importance of relays, new metrics and models have been created so MLB organizations can quantify defensive value. A 2010 article in the Society for American Baseball Research (Reference #4) details the evolution of how fans and organizations quantify run prevention. Both pitching and defense are associated with run prevention, but most

work has been done on the pitching side since the 19th century, but with the development of new metrics beyond errors, fielding percentage, and putouts, the defense portion of run prevention is just as scrutinized as pitching. Some examples brought up in this article include defensive runs saved (DRS), fielding runs, and plus-minus runs saved. Even though the article was written over a decade ago, it's a microcosm in how quantifying run prevention is continuing to evolve.

By combining "Ask Madden" with relay outcome, this fan activity not only involves the action on the field, but also helps fans understand the action on the field by highlighting a crucial but misunderstood component of baseball.

2. Data

The data provided for this project comes from SportsMEDIA Technology, which consists of anonymized MiLB player and ball tracking data from 251 games in an unspecified MiLB season. The data also includes game information as well as the lineups set, which included information regarding if there was a pickoff, the player IDs at each position, and the runners on base on the specific play.

2.1 Constraints and Filtering Rationale

There were certain situational factors that were not included in the data set. For example, the data did not include the number of outs, the outcome of the play, or the score of the game. Also, given the anonymous nature of the data set, a unique player can be represented by multiple player IDs, so it was hard to incorporate player-exclusive components to the model like average moving speed, arm strength, range, etc. (see Appendix F). To determine the outcome of the play, I created certain conditions at the last acquisition of the play to classify if the runner was safe or out (see Appendix C). Even though this may be an imperfect way to classify if a runner was safe or out, I manually looked at each play through a play animation and felt confident in my classification.

The model only considered plays where there was a base hit into the outfield and a potential play at 2nd, 3rd, or home. Under this criterion, it left me with 140 unique plays. These plays were further filtered manually by watching each play in animated GIF format, removing

relays that contained certain nuances or were messy plays (see Appendix B). After further inspection of each relay, I was left with 110 unique relays.

2.2 Exploratory Data Analysis

Both Figure 3 and Figure 4 provide snapshots of the plays in this data set in terms of outcome and where most relays in this dataset end up. Figure 3 shows the slight imbalance in the data set with most of the relays ending with the runner being safe. Figure 4 shows that most of the relays in this data set end up at second base.

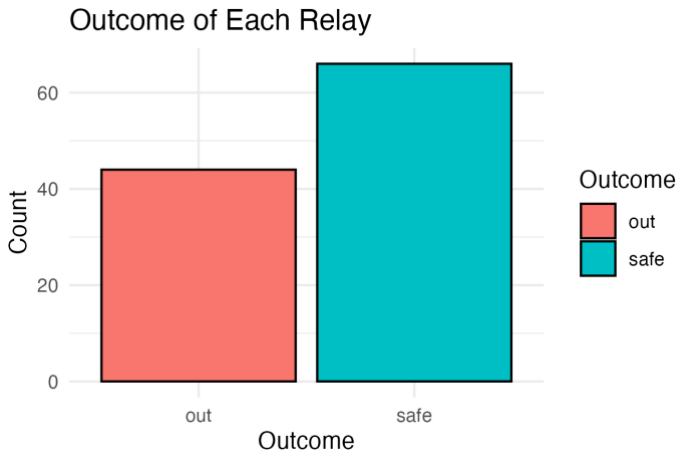


Figure 3: A bar chart showing the number of relays that ended with the runner being safe or out. Out of 110 relays, most relays ended with the runner being safe. This chart also indicates that the data set may be slightly unbalanced, favoring "safe" outcomes.

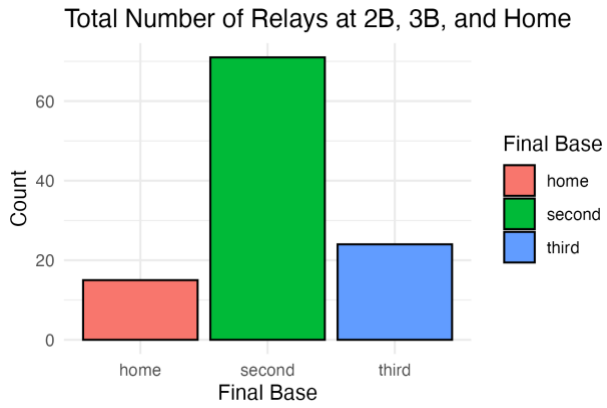


Figure 4: A bar chart showing the number of relays that end at second base, third base, or home. Most relays in this data set end at second base.

3. Methodology and Analysis

The process behind the model boils down to determining the parts of a relay that can be quantified. The key features of the model include the distance between the first acquisition of the ball and its destination, the moving speed of the runner, the velocity on each throw, the time it took for the fielder to get to the ball, the exchange time, if there was a cutoff man involved, the distance of the final throw, and the runner's distance from the base at the final throw. The runner's moving speed and velocity are averages in the model (see Appendix D).

In terms of model selection, I tested four different classification models: standard logistic regression, random forest, a generalized additive model (GAM), and an XGboost model. Ultimately, the model that will be "Ask Madden" is a random forest classification due to its accuracy, recall, and precision scores compared to the other models tested as well as its interpretability (see Appendix E).

In this random forest model, Figure 5 shows the most important features in the model. The most important feature in a successful relay is the speed of the runner, which suggests that the outcome of a relay isn't fully controlled by the defending team. This makes sense because even if all the parts of the relay are perfect, it does not guarantee that the runner will be out. In terms of all the factors that can be controlled by the defense, the most important feature was the exchange time, but other defensive features like the time it took to get the ball, the average velocity on each throw, and the final throw distance share similar importance. Figure 6 shows the model's performance on the test data set, highlighting its accuracy on out and safe plays.

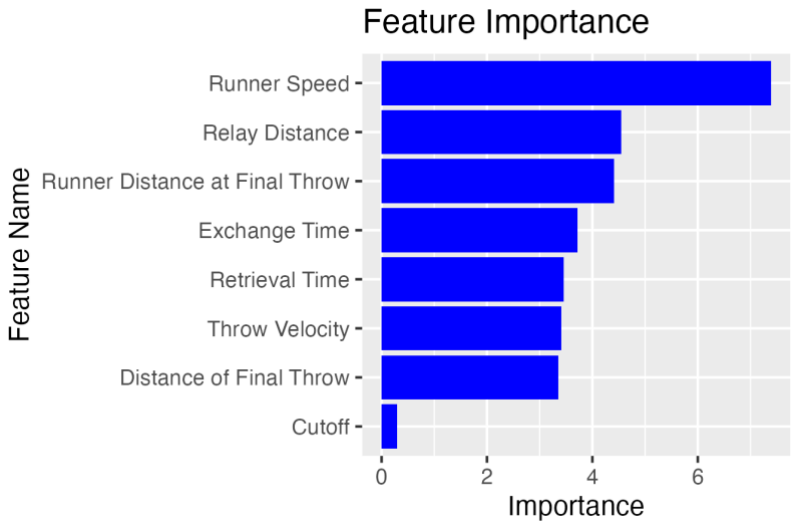


Figure 5: Feature Importance in descending order. The most important feature is the runner's average speed during the play. Defensively, the most important factor that a defense can control is the exchange time, but each defensive variable shares similar importance.

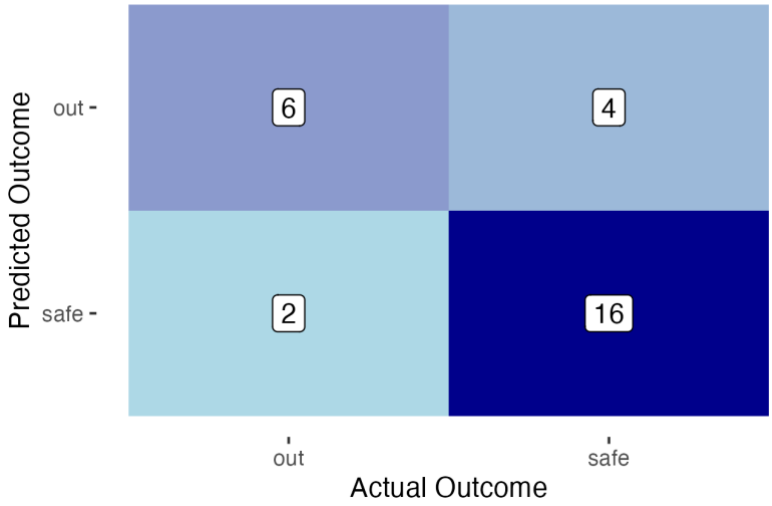


Figure 6: Confusion Matrix for the random forest model on the test data. The model had an 80% success rate on "safe" plays (16/20) and a 75% success rate on "out" plays (6/8).

4. In-Game Implementation

This activity can be implemented at any time during the game, between innings. The activity calls "Ask Madden" as "Ask Maddon", highlighting the versatility teams have in this

activity to cater to their fans. For this demonstration, I used a relay from the 2024 College World Series Final. Since the data for this play is not a part of the competition, any statistic from this play is for demonstration purposes only and does not reflect the true nature of the play. The purpose of this demonstration is to visualize the flow of the activity as well as what fans will see during the events of the activity.

Figure 7 shows how the game is presented visually on a scoreboard. The graphic introducing the activity should be shown during the host's introduction of the participant and the activity (see Appendix A). Figure 8 shows the phase of the play where the last throw is made and the runner's location at that phase. At this phase, there should be a split screen between the fielder and the runner. Figure 9 provides an example output from "Ask Madden". It should include the suggestion as well as a statistic relevant to the play (see Appendix A).



Figure 7: How the game is presented using the scoreboard at Charles Schwab Field.



Figure 8: The split screen between the location of the runner and the location of the fielder at the last throw. This screen should pop up after the ball reaches the last fielder that throws the ball.

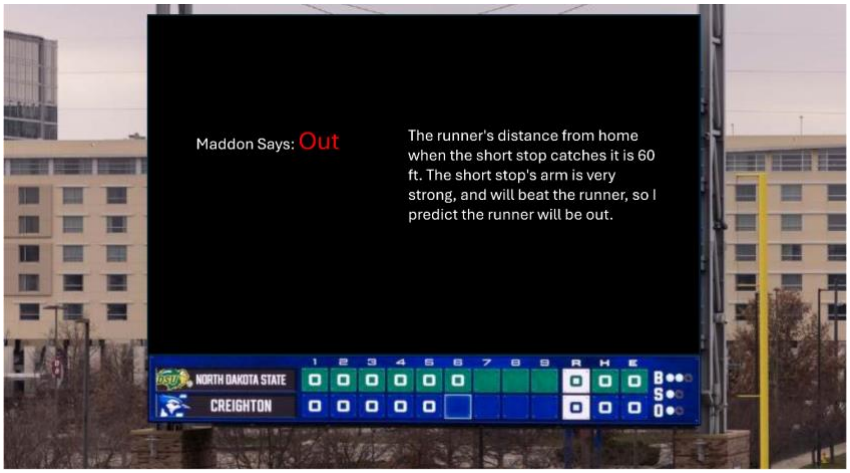


Figure 9: The output from "Ask Madden", renamed "Ask Maddon" for demonstration purposes. The output should state the suggested outcome along with a statistic from that play that led to the suggested outcome.

5. Discussion/Conclusion

The problem with fan interactive activities in ballparks is that they don't encourage active engagement with the action on the field. Creating a safe/out game with the help of "Ask Madden" allows the fan to understand a relatively misunderstood aspect of baseball in a fan-friendly way.

The demonstration shows the versatility of the activity. Teams who implement this activity have the freedom to alter the components of the activity to cater to their fans. For example, teams can change "Ask Madden" to the team's mascot, a club legend, or the team's manager. "Ask Madden" can also be extended to other facets of baseball like home run length, batted ball outcomes, or sacrifice fly plays.

The concept of "Ask Madden" is designed to give the fan the opportunity to engage with the action on the field through a fan-friendly description of the on-field action. As said before, fans understand the game through outcomes, which offer a solid base of understanding, but explaining how the outcome occurred is a different story. In this case, "Ask Madden" allows the fan to understand parts of the relay sequence by describing the action on the field in a fan-friendly way.

6. Acknowledgements

Special thanks to everyone at SMT, especially Dr. Meridith Wills and Billy Fryer for hosting this competition. I would also like to thank everyone that helped me during the process of creating this project whether it was coding, mathematical concepts, or paper writing.

7. References

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igcompany. (2020, August 13). *Madden NFL 12 - PSP Gameplay (4K60fps)*. https://www.youtube.com/watch?v=49af-Dcsfy0

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Reference #5:

NCAA Championships. (2024, June 27). *Tennessee vs. Texas A&M: 2024 Men's College World Series Final Game 3 | FULL REPLAY*. https://www.youtube.com/watch?v=nLREBX34W3M

Reference #6:

Brad Williams. (2025). *New Scoreboards at Charles Schwab Field*. https://growomaha.com/project/new-scoreboards-at-charles-schwab-field/

8. Appendices

Appendix A – Fan Activity Script

Host Duties:

- Must pull a fan aside to ask them to participate
    - The fan can be ANYONE!
- Before the host begins the activity, the host must introduce the contestant by name. However, if the host wants to ask about where they're from, they are allowed to do so. Then the host will describe the activity.

Description and Flow of the Activity

- Host: “We will be playing safe or out! So how this game works is that there will be a close play up on the scoreboard, and you just have to tell us if the runner is safe or out! However, there is a catch! If you need additional help, you are allowed to “Ask Madden” for his advice. If you “Ask Madden”, he will suggest an option based on a statistic from the play!
- **Play starts on the screen, then pauses**
- Host: “Alright [Participant], based on you’ve seen, where the runner is and where the throw is, is the runner safe or out?
- **Give 10 seconds**
- **If participant declines help, skip to when the participant makes the decision**
- ****Participant asks for help from “Ask Madden”****
- Host: “[Name] has asked Madden for advice! What does he suggest?”
- ****Ask Madden makes the suggestion and states some information from the play****
- Host: Repeats the information on the screen
- Host: “With some additional information, do you think the runner is safe or out?”
- ****Give 5 seconds****
- ****Participant makes the decision****
- Host: “[Participant] said [out/safe]! Let’s find out!”
- ****Play runs to the end****
- Host: (If correct) “The runner was indeed [safe/out]! Congratulations [Participant]!”
- Host: (If incorrect) “The runner was not [safe/out].”
- Host: “Thank you so much for participating!”

* Note – the host does not have to say the exact words in this script. The host should be given the freedom to put their own exciting spin on this activity. The demonstration and script do not account for instances where the host can involve the crowd. If someone or an organization is sponsoring this activity, they should be introduced first before the host introduces the participant.

Production Details

- The play in question should start from the pitch before the hitter puts the ball in play. The play should continue to run when the last fielder that throws the ball in the play acquires it.
- Once the ball reaches the last fielder that throws the ball, split the screen to include the runner and where the ball is in the relay sequence.
- If the participant initiates “Ask Madden”, the play needs to be blacked out for “Ask Madden” to give the participant the suggestion and supplemental information.
- After the participant states their answer, the play starts from the last throw and stops when the play ends.

Appendix B – Relay Filtering Rationale

To determine what plays were viewed as potential relays, the play had to be a base hit into the outfield with a play at 2nd, 3rd, or home. Before developing features, I filtered the ball events data set to only include balls in play where the next phase of the play was not a ball acquisition or the play ending. This eliminated catch acquisitions or plays where the data called the play over. For those balls in play, the next acquisition had to come from an outfielder. This would eliminate ground ball plays where a force play at 2nd, 3rd, or home could have occurred.

For a relay to occur, the final base of the runner and the ball needed to match. I needed to classify the final base of the ball and the runner. To classify where the ball ended up, I took the end phase of the play (the last ball acquisition), and calculated the ball’s distance from second base, third base, and home plate using the x and y coordinates of the ball and each base. I then took the minimum distance between those three distances to classify the closest base. However, the minimum distance had to be less than 5 feet from the base. This eliminated plays where the play either ended at first base, plays where there was a cutoff but no throw, or the ball was acquired too far from the base to make a tag. For the runners, I calculated each runner’s distance from second base, third base, and home plate, and then took the minimum distance to classify the end base. The runner had to be within 60 feet from the final base. After creating these classifiers,

the play set was filtered to only include plays where the ball and the runner ended at the same base.

Despite these conditions, there were many noisy plays that were considered relays. For example, Figure 10 shows a play where the initial intention was to get the runner out at home, but the play ends at 2nd base when the first baseman cuts the ball off to go to 2nd. Figure 11 shows an example where the play ends at 2nd base, but the hitter intentionally gets in a pickle to allow the runner to head home. So, to check the quality of the relay set, I watched every play in the relay set to determine if it was a relay. I removed plays that either had a rundown play, a cutoff on a play to home that ended up at either 2nd or 3rd, or plays where an overthrow happens, and the play ends somewhere else.

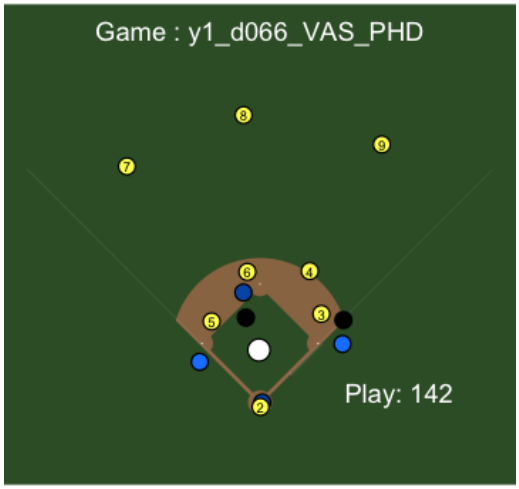


Figure 10: (Link to GIF) An example of a play heading home, but the play ends at 2nd base after a first base cutoff

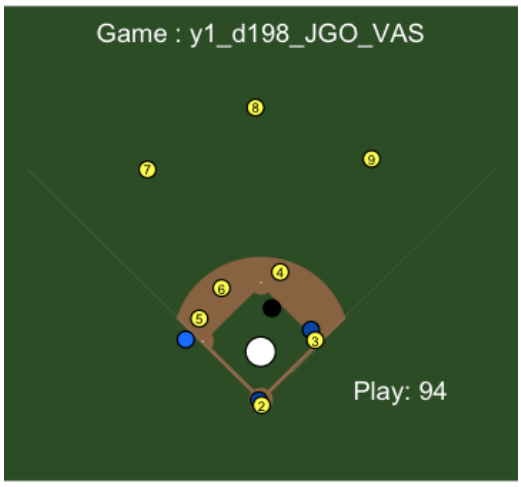


Figure 11: (Link to GIF) An example of a play where the hitter intentionally gets in a pickle to allow the runner to head home.

Appendix C – Determining Relay Outcome

As stated in section 2.1, there is no event code or variable that mentions the outcome of the play, so I had to classify whether a player was safe or out. To achieve this, I calculated the distance between the runner and the final base when the ball reaches the final base. I considered the runner safe if the distance between the runner and the final base was less than 7 feet to account for the runner's slide and the time to tag the runner. However, another constraint is that there isn't a great way to account for other nuances associated with the play such as throw quality and how far the ball pulls the player away from the base, which affects the player's ability to tag the runner. However, I felt comfortable with these conditions by watching the play animations and manually checking the plays using the lineups data set.

Appendix D – Feature Engineering

In determining the features of this model, I wanted to include features that reflected the components of the relay. This included the time to retrieve the ball, exchange time, the relay distance, the speed of the runner, the velocity on each throw, the distance between the runner and the final base at the last throw, the distance between the ball and the base on the final throw, and determining if there was a cutoff on the play.

To calculate the retrieval time and exchange time, I filtered the set to only include the ball in play and its subsequent throws and acquisitions. For retrieval time, I calculated the difference between the time when the ball was in play and when the player acquired the ball. For exchange time, I calculated the time difference between the acquisition of the ball and the time when the player releases the ball.

For velocity on each throw, I calculated the total distance between the throw and acquisition and divided it by the time difference between those two phases. With this feature, I ignored the effect of ball bounces, assuming that the change in velocity from those bounces had a minimal effect on the overall pace of the ball.

To determine if there was a cutoff man on the relay, I used the ball events data set to determine if at any time during the play sequence, an infielder acquired the ball from an outfield throw.

For calculating the total relay distance, I created a feature that calculated the ball's distance from the ball's final destination at each phase of play. I then took the max value of that variable to determine the distance between the first acquisition and the last acquisition. To find the runner's distance from the end at the last throw and the ball's distance from the end at the last throw, I had to determine if there was a cutoff involved. For plays without a cutoff, the ball's distance from the end is calculated between the location of the ball and the base when the outfielder makes the throw. The runner's distance from the end is the runner's location from the base when the outfielder makes the throw. For plays with a cutoff, the ball's distance from the end is calculated between where the cutoff man makes the throw and its final location, and the runner's distance from the end is calculated between the runner‘s location and the base when the cutoff man makes the throw.

Runner speed was calculated at each phase of the play by calculating the distance covered by the runner between each phase, and dividing it by the time difference between each phase. For this model, the runner's speed and throw velocity are aggregate means throughout the play.

Appendix E – Model and Model Assumptions

For the modeling process, I made a series of assumptions for each model. One assumption is that the model does not account for throw quality or nuance associated with the tag. The model will assume that if the ball reaches the base before the runner is 7 feet away from base, the runner is out regardless of how good or bad the throw is. I'm making this assumption because there isn't any video to detect certain nuances like the time between the acquisition and the tag as well as how good the throw is. Another assumption is that the game situation is independent of each relay situation. As mentioned in section 2.1, the data is missing the score, the number of outs in the inning, and a team's defensive positioning. Since the data is anonymized, park dimensions are not known because I don't know the team's ballpark, so it's hard to classify if the outfield is playing deep or not. Finally, the model assumes that the outcome is independent from where the relay ends up. For example, the model assumes that the change in probability on a relay to second versus the change in probability on a relay to third base is insignificant. Despite these concerns, the model should still provide insight into what components of a relay are significant in the outcome of a relay. Finally, the model will assume that deviations in runner speed and velocity on each throw aren't impactful enough to affect the model's ability to predict the outcome of the relay. For the models tested, it's impossible to account for all phases of the play, so both runner speed and throw velocity are aggregated means. Even though the model can't account for the dynamics of the play, using aggregated means in throw velocity and runner speed can provide a good snapshot of the relay.

For the model, I tested four different classification models. I started with a logistic regression model, which provides the most interpretability, but it can't capture complex relationships between the features and the response compared to other models. The second model I tested was a generalized additive model, which allows for the inclusion of linear and non-linear terms, providing flexibility for different types of data. Despite the added flexibility to the model, a generalized additive model doesn't capture complex relationships as well as other black-box models and loses interpretability. The third model I tested was a random forest model, which provides the flexibility associated with a generalized additive model, but uses many decision trees to determine response. A similar model is the XGboost model, which has all the perks of a random forest model, but the decision trees learn from each other, correcting the errors of

previous trees. Even though both XGboost and random forest offer the greatest versatility out of all the classification models and are typically more accurate, they are both not very interpretable given their model complexity, and the models are very prone to overfitting.

The metrics that differentiated each model were the model's accuracy, precision, and recall. Accuracy summarizes how many plays the model predicted correctly out of the total number of plays. Precision refers to the model's accuracy on plays that ended with the runner being out. Recall refers to the proportion of the number of plays the model predicted to be out among all plays that ended with the runner being out.

Figure 12 shows the performance of each model in terms of the model's accuracy, recall, and precision. As said in section 3, "Ask Madden" will be a random forest model due to its performance compared to the other models. The worst model was the generalized additive model, as it had the lowest accuracy, precision, and recall. The second worst was the XGboost model. The low sample size probably played a role in why these two models were not performing well. Also, the data wasn't complex enough to make these models stand out. Even though the random forest model and the basic logistic regression model were relatively close in terms of accuracy, precision, and recall, I decided to use the random forest model instead because it's able to capture complex relationships between its features without fully compromising the interpretability of the model.

model_type	test_accuracy	test_recall	test_precision
log_model	71.42857	50.0	50.00000
random_forest	78.57143	75.0	60.00000
gam	25.00000	25.0	11.76471
XGboost	67.85714	62.5	45.45455

Figure 12: The accuracy, precision, and recall of each model. The random forest model had the highest accuracy, precision, and recall value out of all the models tested on the test data.

Appendix F – Future Work

The model presented in this activity is a foundation for further refinement of the model. An obvious solution is to have more data that could train and test the model. Even though the relatively low sample size reinforces the idea that relays are miniscule in the grand scheme of the game, having more relays would provide a better gauge of model performance. In terms of additional features, one potential improvement is incorporating features accounting for player ability and past performance. As stated before, having measures of player ability and performance wasn't possible given that not every player had one unique ID to preserve the anonymous nature of the data. By adding features that describe player ability of each player in the sequence, it allows the model to account for a player's ability to affect the outcome. Another way the model can be refined is by adding situational-based features like defensive alignment, number of outs, and the score of the game. The addition of situational-based features may allow the model to account for how aggressive or passive the defense can be regarding potential plays. Some other features that could be accounted for in the model are park-related features as not every ballpark is the same, which could affect the outcome of the play.

In terms of the model itself, another solution could be to use a model like a win-probability model. By having a model that can provide probabilities at each phase of the play instead of aggregating certain features, the model can capture certain nuances of a relay that aggregated data cannot.